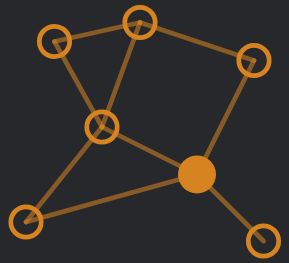
PERYCLES · WP5

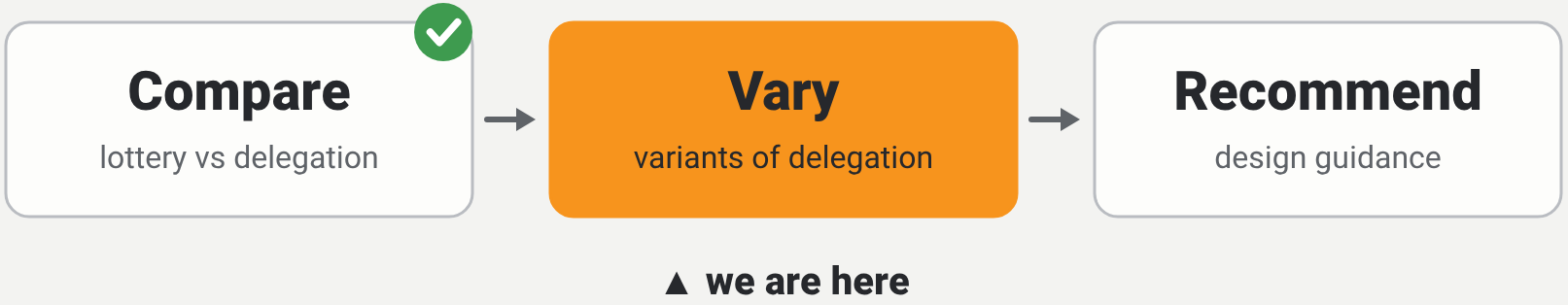
Variants of Liquid Representation



Nimrod Talmon · with Eyal Briman and Davide Grossi

Ben-Gurion University of the Negev · Rome, December 2026

WP5: where we are



Recall · D5.1

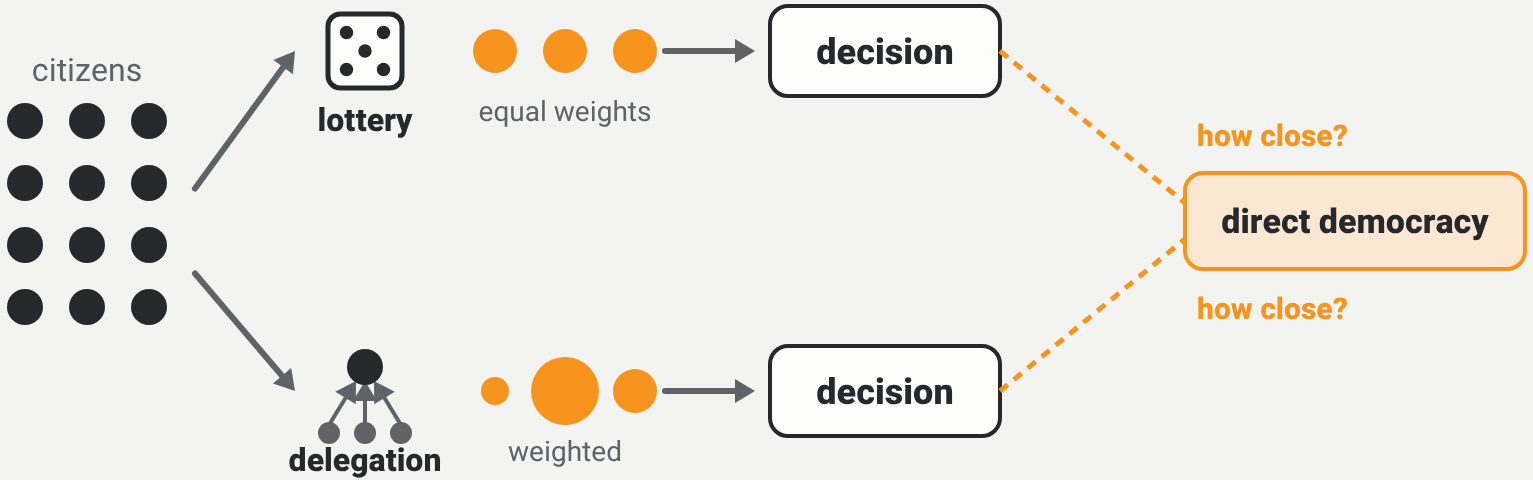
Lottery or delegation?

Paper: “Lot or Proxy? Representation in Sortition and Liquid Democracy”

with Eyal Briman and Davide Grossi · 🤞 AAMAS 2027

Recall · D5.1

A committee by lottery, or by delegation?



Which is more accurate?

It depends on the population

Polarized



lottery more accurate

Moderate



delegation more accurate

New: more simulations; same main message, in more detail

e.g., in some populations, committee size matters too

Small committee



lottery more accurate

Large committee



delegation more accurate

Variant 1

Topic-based delegation

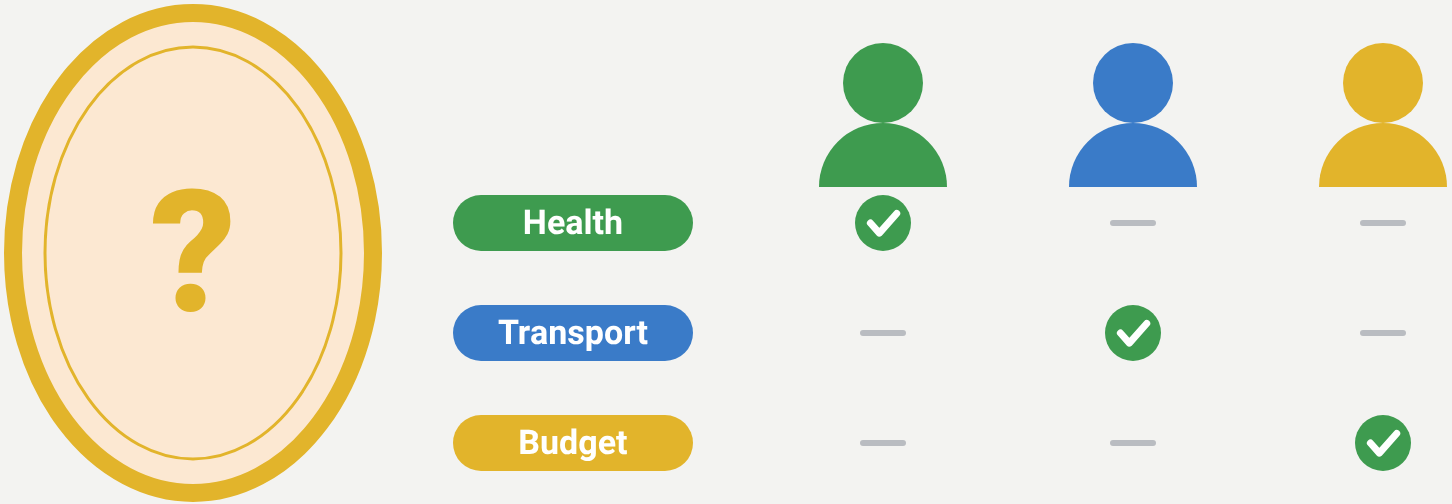
Paper: “Topic Bundling for Epistemic Committee Design”

with Davide Grossi and Eyal Briman · 🤞 AAMAS 2027

Variant 1 · the problem

Mirror, mirror: who is my best delegate?

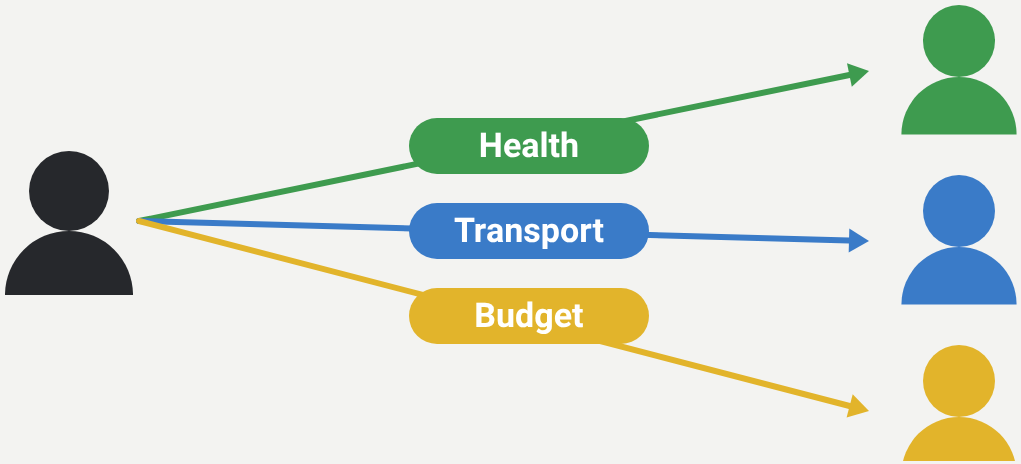
With one delegation for everything, no single delegate may be best on all topics



Variant 1 · the approach

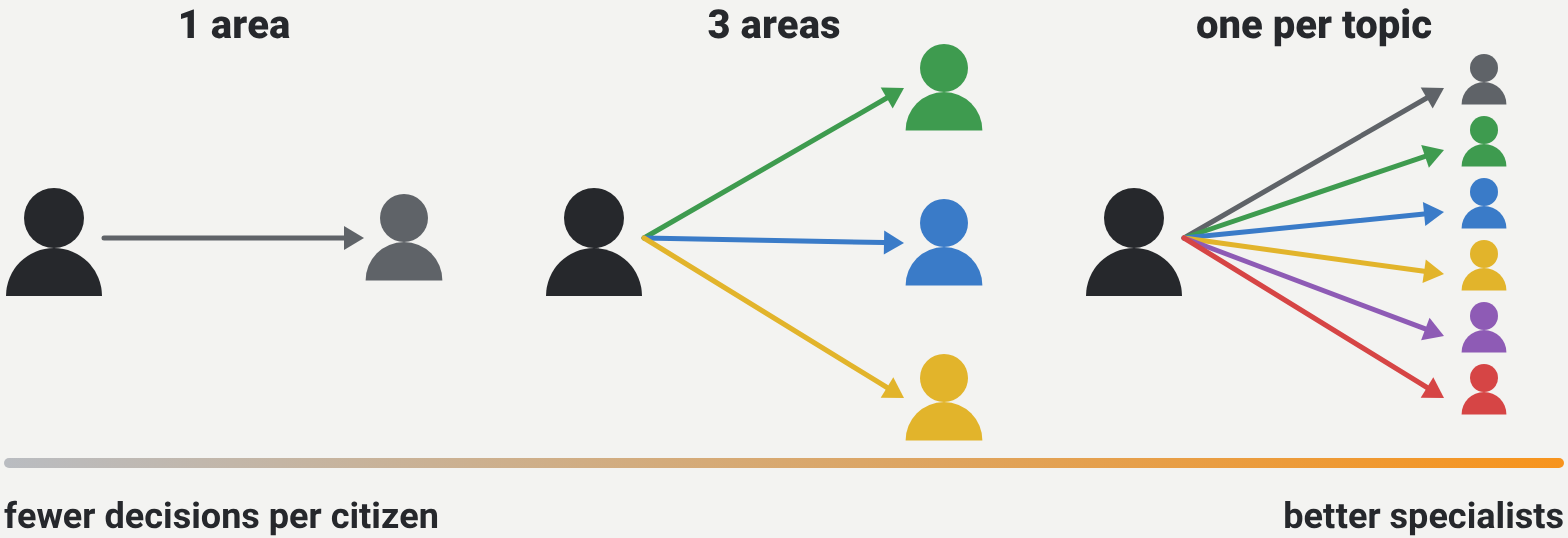
Delegate per area

Our question: how fine should the areas be?



Variant 1

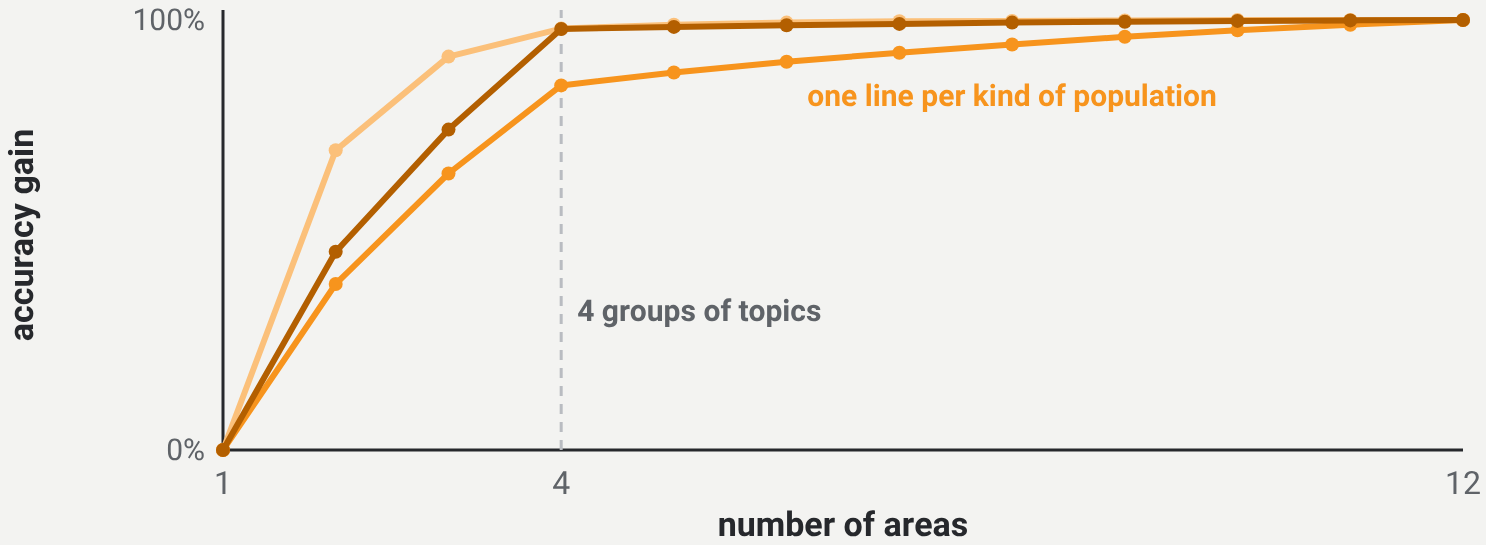
How many areas?



Variant 1 · what we show

A few areas go a long way

More areas never lower accuracy, but the gain levels off (experiments, 12 topics in 4 groups)



Variant 2

Sortition on top of delegation

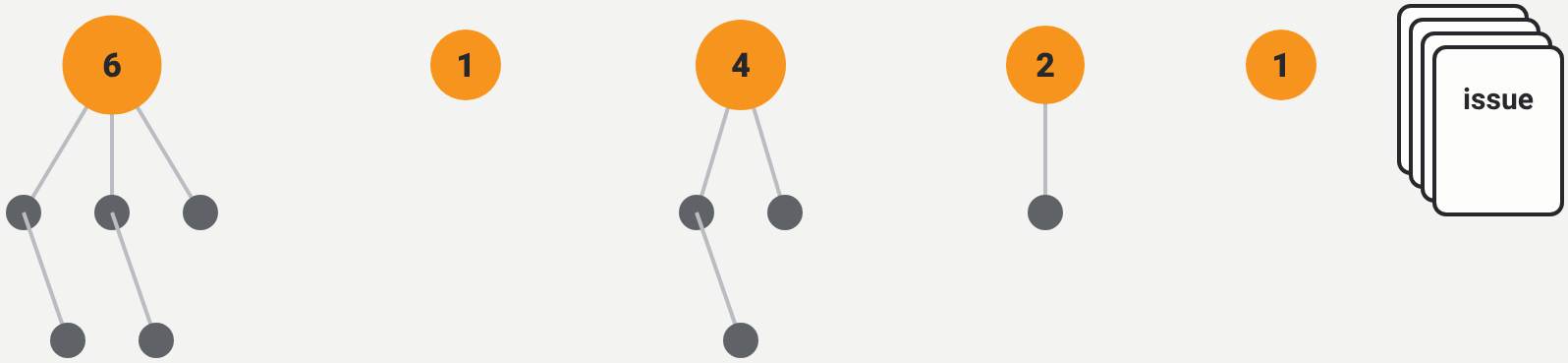
Paper: Sampled liquid democracy

with Eyal Briman ·  in the oven

Variant 2 · the problem

Representatives must vote on everything

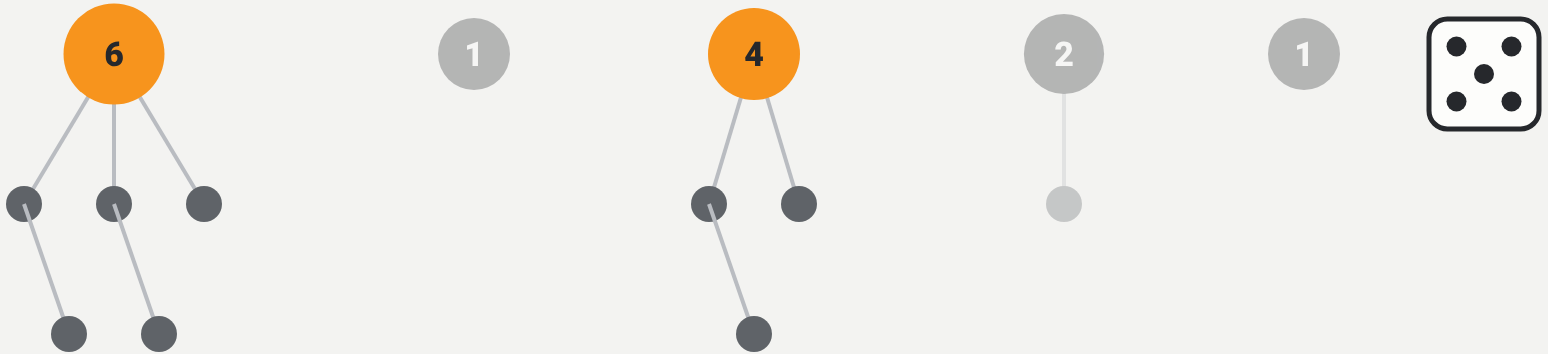
Some may be busy, absent, or vote without attention



Variant 2 · the approach

Draw a small panel

Large representatives are drawn more often; drawn votes are scaled up to stand for the others



Like a poll: a random sample tracks the whole

≈ 160

panel members match a 60/40 full vote at least 95% of the time

whatever the population size

Variant 2 · two knobs

Panel size, and how much delegations count

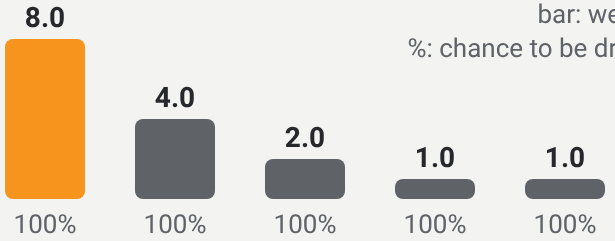
Example: representatives hold 8, 4, 2, 1, 1 citizens and are right 80, 70, 65, 60, 60% of the time

Panel size 5 (everyone) (1) ◀ ▶ (2)

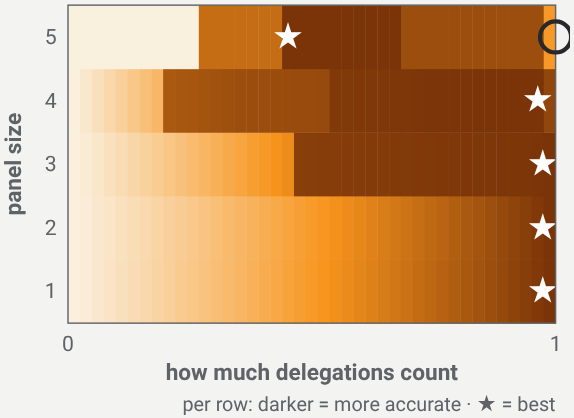
How much delegations count 1.00 (3) ◀ ▶ (4)

0: everyone counts 1 — 1: full delegation weight

bar: weight
%: chance to be drawn



Accuracy 81.0%

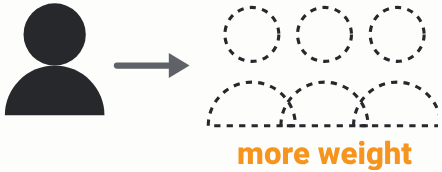


per row: darker = more accurate · ★ = best

Variant 2

The price of counting delegations less

~~1 person, 1 vote~~



Fake identities pay



Small accuracy gains

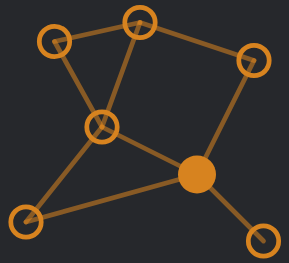
Towards D5.3

Let's discuss

1. What suggestions do we want to give?
2. What would we want implemented?
3. Other variants worth studying?

PERYCLES · WP5

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